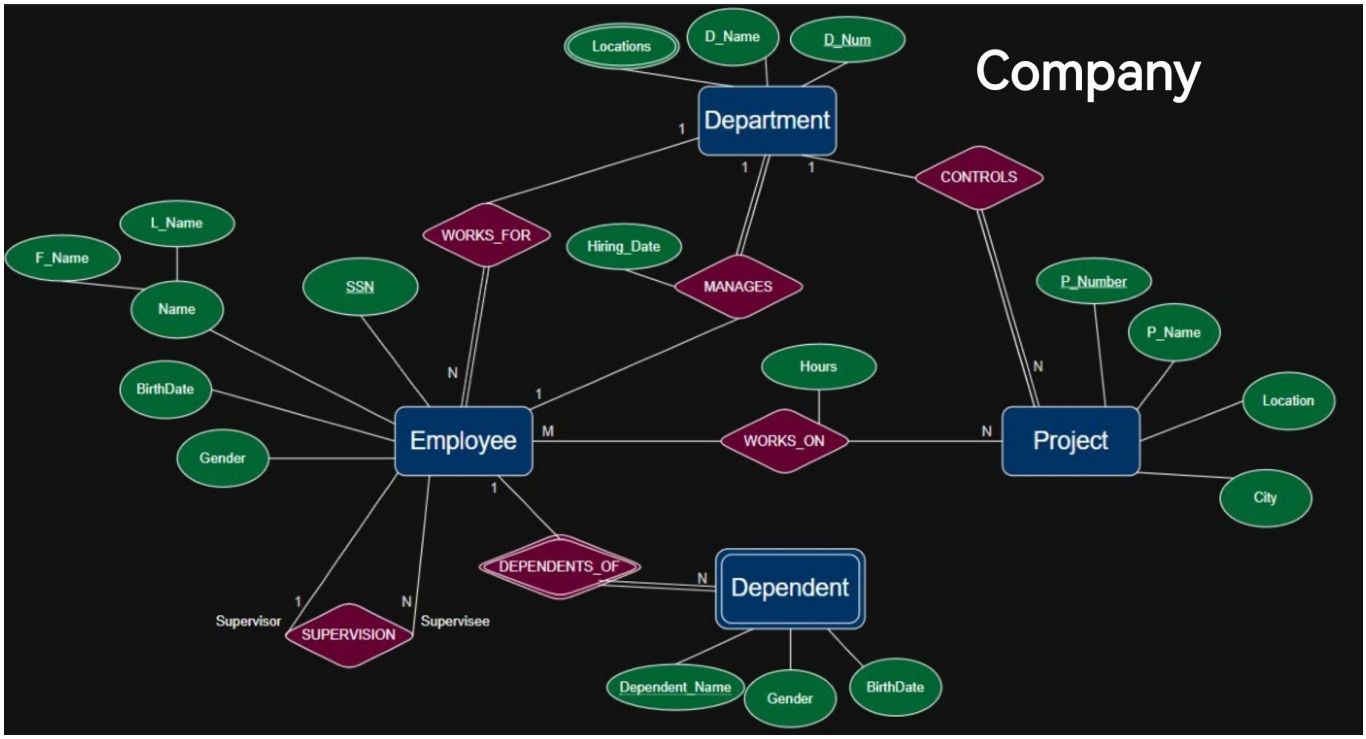
Entity Relationship Diagram & Relational Mapping

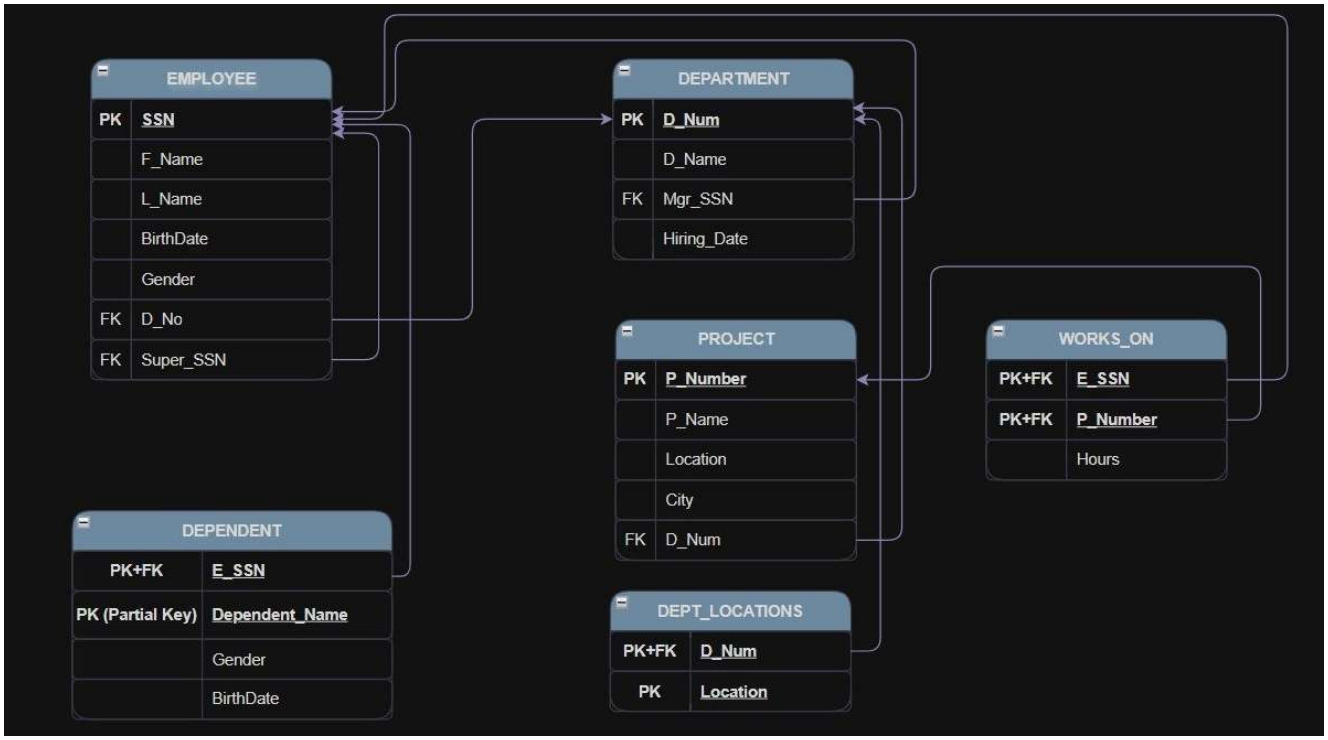
Database Systems

Eng /Reem Ayman

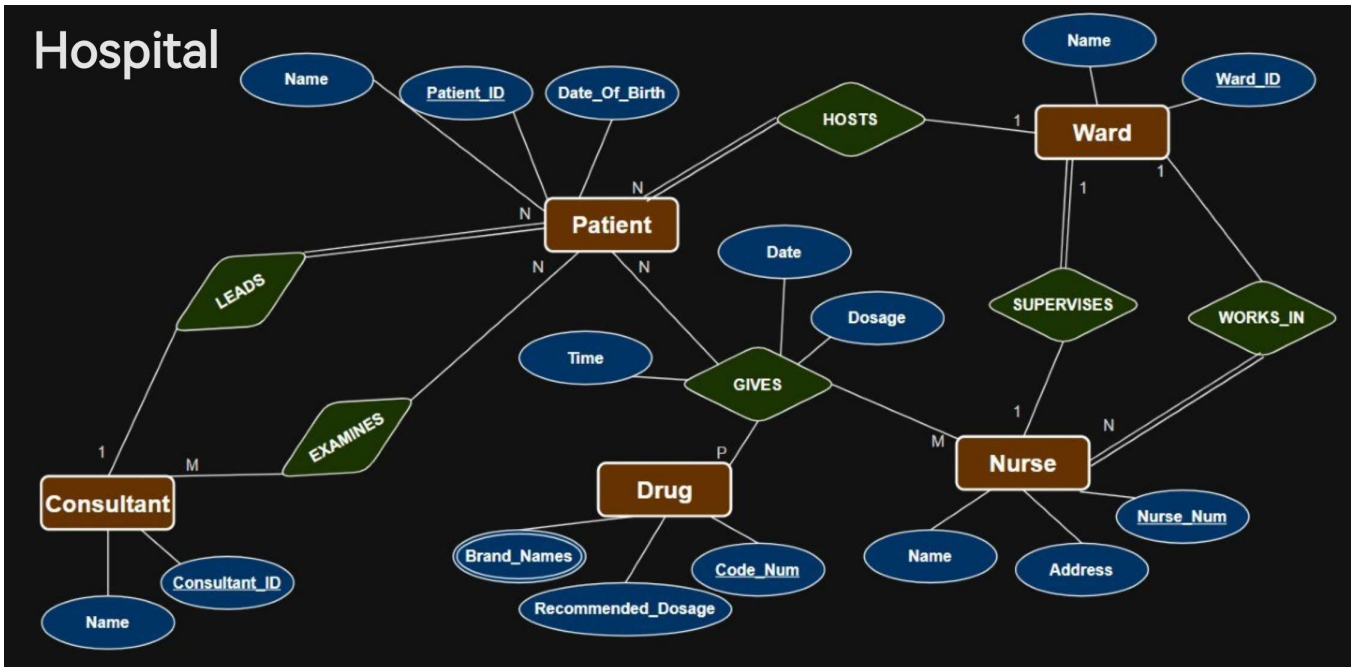
ERD


Company
Locations
D_Name
D_Num
Department
CONTROLS
WORKS_FOR
Hiring_Date
MANAGES
L_Name
F_Name
Name
SSN
BirthDate
Gender
Employee
Hours
WORKS_ON
Project
P_Number
P_Name
Location
City
DEPENDENTS_OF
Dependent
Supervisor
SUPERVISION
Supervisee
Dependent_Name
Gender
BirthDate

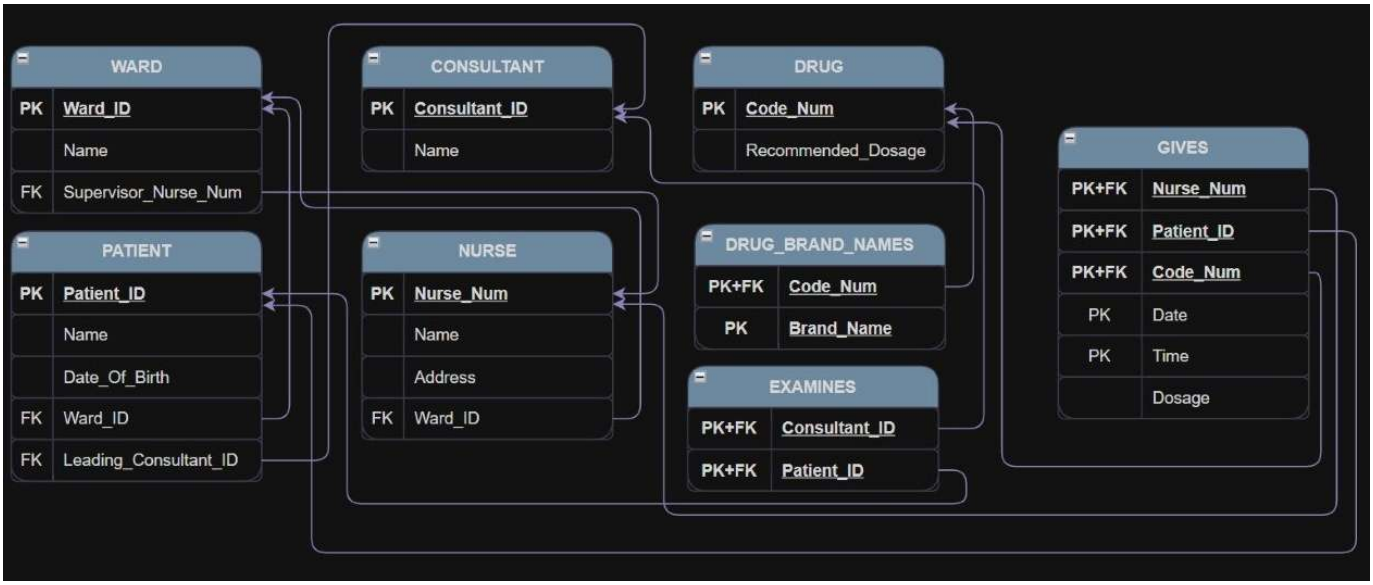
Mapping


EMPLOYEE
PK SSN
F_Name
L_Name
BirthDate
Gender
FK D_No
FK Super_SSN
DEPARTMENT
PK D_Num
D_Name
FK Mgr_SSN
Hiring_Date
PROJECT
PK P_Number
P_Name
Location
City
FK D_Num
WORKS_ON
PK+FK E_SSN
PK+FK P_Number
Hours
DEPENDENT
PK+FK E_SSN
PK (Partial Key) Dependent_Name
Gender
BirthDate
DEPT_LOCATIONS
PK+FK D_Num
PK Location

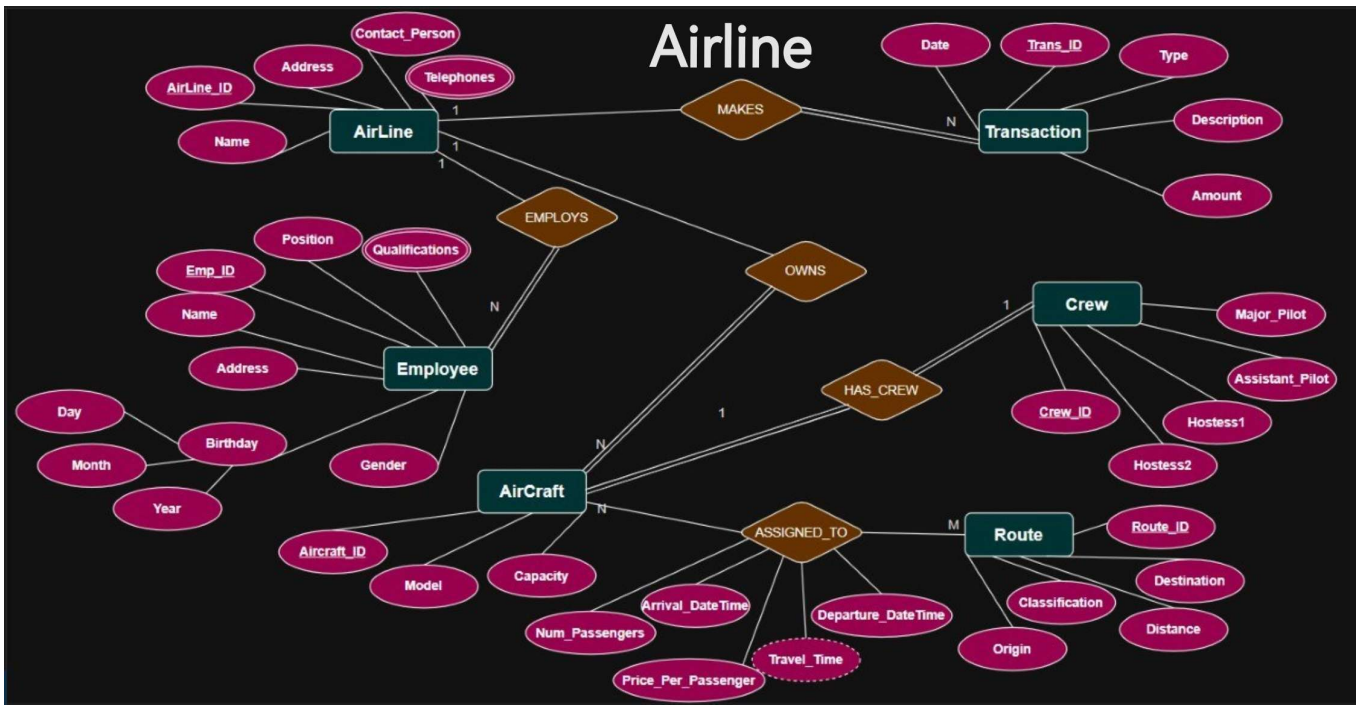
ERD


Hospital
Name
Patient_ID
Date_Of_Birth
HOSTS
Name
Ward_ID
Ward
Patient
LEADS
Date
Dosage
SUPERVISES
WORKS_IN
Time
GIVES
EXAMINES
Consultant
Drug
Nurse
Nurse_Num
Brand_Names
Code_Num
Name
Address
Consultant_ID
Name
Recommended_Dosage

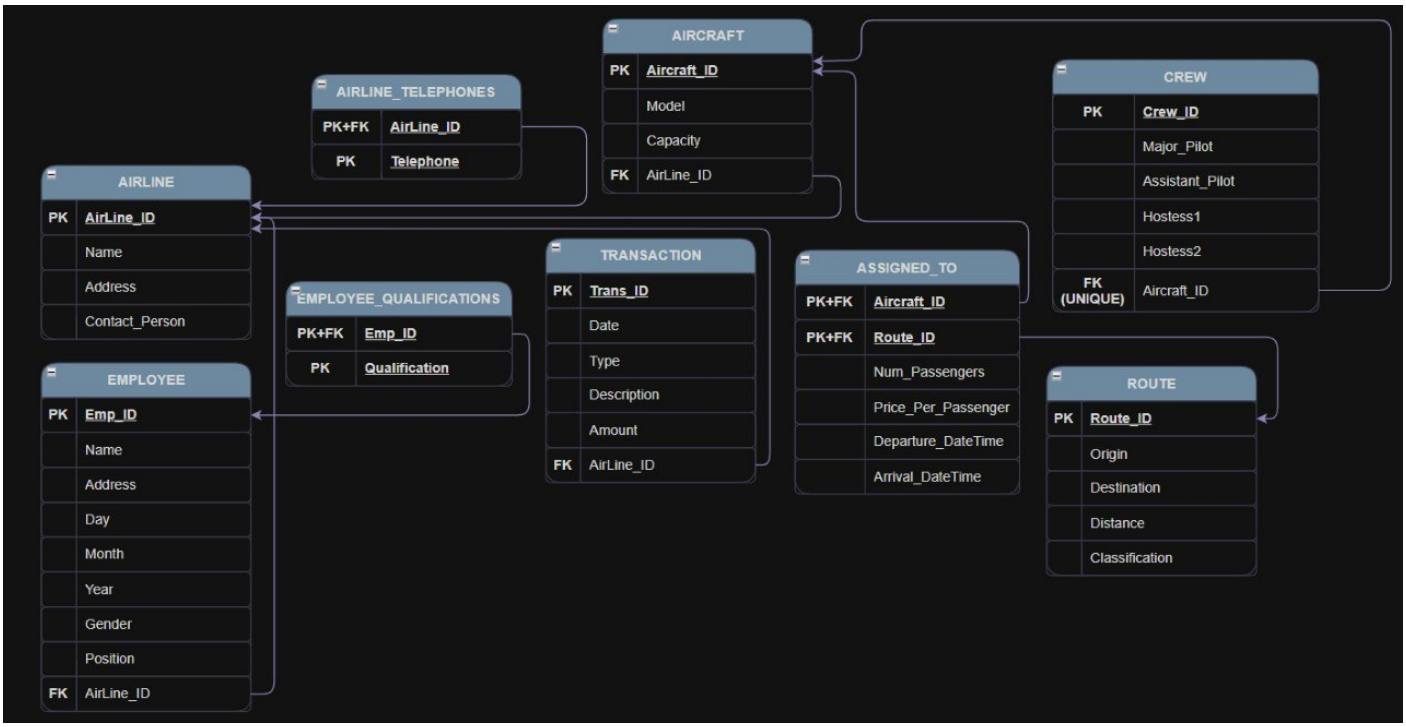
Mapping


WARD
PK Ward_ID
Name
FK Supervisor_Nurse_Num
CONSULTANT
PK Consultant_ID
Name
DRUG
PK Code_Num
Recommended_Dosage
GIVES
PK+FK Nurse_Num
PK+FK Patient_ID
PK+FK Code_Num
PK Date
PK Time
Dosage
PATIENT
PK Patient_ID
Name
Date_Of_Birth
FK Ward_ID
FK Leading_Consultant_ID
NURSE
PK Nurse_Num
Name
Address
FK Ward_ID
DRUG_BRAND_NAMES
PK+FK Code_Num
PK Brand_Name
EXAMINES
PK+FK Consultant_ID
PK+FK Patient_ID

ERD


Airline
Contact_Person
Address
Telephones
AirLine_ID
Name
AirLine
MAKES
Date
Trans_ID
Type
Description
Amount
Transaction
EMPLOYS
Position
Qualifications
Emp_ID
Name
Address
Employee
OWNS
Crew
Major_Pilot
Assistant_Pilot
Hostess1
Hostess2
Crew_ID
HAS_CREW
Day
Month
Year
Birthday
Gender
AirCraft
Aircraft_ID
Model
Capacity
ASSIGNED_TO
Route
Route_ID
Destination
Distance
Origin
Classification
Arrival_DateTime
Departure_DateTime
Num_Passengers
Travel_Time
Price_Per_Passenger

Mapping


AIRLINE
PK AirLine_ID
Name
Address
Contact_Person
AIRLINE_TELEPHONES
PK+FK AirLine_ID
PK Telephone
AIRCRAFT
PK Aircraft_ID
Model
Capacity
FK AirLine_ID
CREW
PK Crew_ID
Major_Pilot
Assistant_Pilot
Hostess1
Hostess2
FK (UNIQUE) Aircraft_ID
EMPLOYEE_QUALIFICATIONS
PK+FK Emp_ID
PK Qualification
TRANSACTION
PK Trans_ID
Date
Type
Description
Amount
FK AirLine_ID
ASSIGNED_TO
PK+FK Aircraft_ID
PK+FK Route_ID
Num_Passengers
Price_Per_Passenger
Departure_DateTime
Arrival_DateTime
ROUTE
PK Route_ID
Origin
Destination
Distance
Classification
EMPLOYEE
PK Emp_ID
Name
Address
Day
Month
Year
Gender
Position
FK AirLine_ID